

Digital Electronics Question and Answers

Arithmetic Circuits

Topic 4

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Arithmetic Circuits

1.) Introduction of Arithmetic Operation

1. The basic building blocks of the arithmetic unit in digital computers are _____

- a) Subtractors
- b) Adders
- c) Multiplexer
- d) Comparator

Answer: b

Explanation: The basic building blocks of the arithmetic unit in digital computers are adders. Since a parallel adder is constructed with a number of full-adder circuits connected in cascade. By controlling the data inputs to the parallel adder, it is possible to obtain different types of arithmetic operations.

2. A digital system consists of _____ types of circuits.

- a) 2
- b) 3
- c) 4
- d) 5

Answer: a

Explanation: A digital system consists of two types of circuits and these are a combinational and sequential logic circuit. Combinational circuits are the ones which do not depend on previous inputs while Sequential circuits depend on past inputs.

3. In a combinational circuit, the output at any time depends only on the _____ at that time.

- a) Voltage
- b) Intermediate values
- c) Input values
- d) Clock pulses

Answer: c

Explanation: In a combinational circuit, the output at any time depends only on the input values at that time and not on past or intermediate values.

4. In a sequential circuit, the output at any time depends only on the input values at that time.

- a) Past output values
- b) Intermediate values
- c) Both past output and present input
- d) Present input values

Answer: c

Explanation: In a sequential circuit, the output at any time depends on the present input values as well as past output values. It also depends on clock pulses depending on whether it's synchronous or asynchronous sequential circuits.

5. Procedure for the design of combinational circuits are:

- A. From the word description of the problem, identify the inputs and outputs and draw a block diagram.
- B. Draw the truth table such that it completely describes the operation of the circuit for different combinations of inputs.
- C. Simplify the switching expression(s) for the output(s).
- D. Implement the simplified expression using logic gates.
- E. Write down the switching expression(s) for the output(s).

- a) B, C, D, E, A
- b) A, D, E, B, C
- c) A, B, E, C, D
- d) B, A, E, C, D

Answer: c

Explanation: Combinational circuits are the ones which do not depend on previous inputs and depends only on the present values. The given arrangement A, B, E, C, D is the right sequence for the designing of the combinational circuits.

6. All logic operations can be obtained by means of _____

- a) AND and NAND operations
- b) OR and NOR operations
- c) OR and NOT operations
- d) NAND and NOR operations

Answer: d

Explanation: Since the logic gates NOR and NAND are known as universal logic gates, therefore it can be used to design all the three basic gates AND, OR and NOT. Thus, it means that any operations can be obtained by implementation of these gates.

7. The design of an ALU is based on _____

- a) Sequential logic
- b) Combinational logic
- c) Multiplexing
- d) De-Multiplexing

Answer: b

Explanation: The design of an ALU is based on combinational logic. Because the unit has a regular pattern, it can be broken into identical stages connected in cascade through carries.

8. If the two numbers are unsigned, the bit conditions of interest are the _____ carry and a possible _____ result.

- a) Input, zero
- b) Output, one
- c) Input, one
- d) Output, zero

Answer: d

Explanation: If the two numbers are unsigned, the bit conditions of interest are the output carry and a possible zero result.

9. If the two numbers include a sign bit in the highest order position, the bit conditions of interest are the sign of the result, a zero indication and _____

- a) An underflow condition
- b) A neutral condition
- c) An overflow condition
- d) One indication

Answer: c

Explanation: If the two numbers include a sign bit in the highest order position, the bit conditions of interest are the sign of the result, a zero indication and an overflow condition.

10. The flag bits in an ALU is defined as _____

- a) The total number of registers
- b) The status bit conditions
- c) The total number of control lines
- d) All of the Mentioned

Answer: b

Explanation: In an ALU, status bit conditions are sometimes called condition code bits or flag bits. It is so called because they tend to represent the status of the respect flags after any operation.

2.) Half Adder & Full Adder

1. In parts of the processor, adders are used to calculate _____

- a) Addresses
- b) Table indices
- c) Increment and decrement operators
- d) All of the Mentioned

Answer: d

Explanation: Adders are used to perform the operation of addition. Thus, in parts of the processor, adders are used to calculate addresses, table indices, increment and decrement operators, and similar operations.

2. Total number of inputs in a half adder is _____

- a) 2
- b) 3
- c) 4
- d) 1

Answer: a

Explanation: Total number of inputs in a half adder is two. Since an EXOR gates has 2 inputs and carry is connected with the input of EXOR gates. The output of half-adder is also 2, them being, SUM and CARRY. The output of EXOR gives SUM and that of AND gives carry.

3. In which operation carry is obtained?

- a) Subtraction
- b) Addition
- c) Multiplication
- d) Both addition and subtraction

Answer: b

Explanation: In addition, carry is obtained. For example: $101 + 111 = 100$; in this example carry is obtained after 1st addition (i.e. $1 + 1 = 10$). In subtraction, borrow is obtained. Like, $0 - 1 = 1$ (borrow 1).

4. If A and B are the inputs of a half adder, the sum is given by _____

- a) A AND B
- b) A OR B
- c) A XOR B
- d) A EX-NOR B

Answer: c

Explanation: If A and B are the inputs of a half adder, the sum is given by A XOR B, while the carry is given by A AND B.

5. If A and B are the inputs of a half adder, the carry is given by _____

- a) A AND B
- b) A OR B
- c) A XOR B
- d) A EX-NOR B

Answer: a

Explanation: If A and B are the inputs of a half adder, the carry is given by: A(AND)B, while the sum is given by A XOR B.

6. Half-adders have a major limitation in that they cannot _____

- a) Accept a carry bit from a present stage
- b) Accept a carry bit from a next stage
- c) Accept a carry bit from a previous stage
- d) Accept a carry bit from the following stages

Answer: c

Explanation: Half-adders have a major limitation in that they cannot accept a carry bit from a previous stage, meaning that they cannot be chained together to add multi-bit numbers. However, the two output bits of a half-adder can also represent the result $A+B=3$ as sum and carry both being high.

7. The difference between half adder and full adder is _____

- a) Half adder has two inputs while full adder has four inputs
- b) Half adder has one output while full adder has two outputs
- c) Half adder has two inputs while full adder has three inputs
- d) All of the Mentioned

Answer: c

Explanation: Half adder has two inputs while full adder has three inputs; this is the difference between them, while both have two outputs SUM and CARRY.

8. If A, B and C are the inputs of a full adder then the sum is given by _____

- a) A AND B AND C
- b) A OR B AND C
- c) A XOR B XOR C
- d) A OR B OR C

Answer: c

Explanation: If A, B and C are the inputs of a full adder then the sum is given by A XOR B XOR C.

9. If A, B and C are the inputs of a full adder then the carry is given by _____

- a) $A \text{ AND } B \text{ OR } (A \text{ OR } B) \text{ AND } C$
- b) $A \text{ OR } B \text{ OR } (A \text{ AND } B) C$
- c) $(A \text{ AND } B) \text{ OR } (A \text{ AND } B)C$
- d) $A \text{ XOR } B \text{ XOR } (A \text{ XOR } B) \text{ AND } C$

Answer: a

Explanation: If A, B and C are the inputs of a full adder then the carry is given by $A \text{ AND } B \text{ OR } (A \text{ OR } B) \text{ AND } C$, which is equivalent to $(A \text{ AND } B) \text{ OR } (B \text{ AND } C) \text{ OR } (C \text{ AND } A)$.

10. How many AND, OR and EXOR gates are required for the configuration of full adder?

- a) 1, 2, 2
- b) 2, 1, 2
- c) 3, 1, 2
- d) 4, 0, 1

Answer: b

Explanation: There are 2 AND, 1 OR and 2 EXOR gates required for the configuration of full adder, provided using half adder. Otherwise, configuration of full adder would require 3 AND, 2 OR and 2 EXOR.

3.) Half & Full Subtractor

1. Half subtractor is used to perform subtraction of _____

- a) 2 bits
- b) 3 bits
- c) 4 bits
- d) 5 bits

Answer: a

Explanation: Half subtractor is a combinational circuit which is used to perform subtraction of two bits, namely minuend and subtrahend and produces two outputs, borrow and difference.

2. For subtracting 1 from 0, we use to take a _____ from neighbouring bits.

- a) Carry
- b) Borrow
- c) Input
- d) Output

Answer: b

Explanation: For subtracting 1 from 0, we use to take a borrow from neighbouring bits because carry is taken into consideration during addition process.

3. How many outputs are required for the implementation of a subtractor?

- a) 1
- b) 2
- c) 3
- d) 4

Answer: b

Explanation: There are two outputs required for the implementation of a subtractor. One for the difference and another for borrow.

4. Let the input of a subtractor is A and B then what the output will be if $A = B$?

- a) 0
- b) 1
- c) A
- d) B

Answer: a

Explanation: The output for $A = B$ will be 0. If $A = B$, it means that $A = B = 0$ or $A = B = 1$. In both of the situation subtractor gives 0 as the output.

5. Let A and B is the input of a subtractor then the output will be _____

- a) $A \text{ XOR } B$
- b) $A \text{ AND } B$
- c) $A \text{ OR } B$
- d) $A \text{ EXNOR } B$

Answer: a

Explanation: The subtractor has two outputs BORROW and DIFFERENCE. Since the difference output of a subtractor is given by $AB' + BA'$ and this is the output of a XOR gate. So, the final difference output is $AB' + BA'$.

6. Let A and B is the input of a subtractor then the borrow will be _____

- a) $A \text{ AND } B'$
- b) $A' \text{ AND } B$
- c) $A \text{ OR } B$
- d) $A \text{ AND } B$

Answer: b

Explanation: The borrow of a subtractor is received through AND gate whose one input is inverted. On that basis the borrow will be $(A' \text{ AND } B)$.

7. What does minuend and subtrahend denotes in a subtractor?

- a) Their corresponding bits of input

- b) Its outputs
- c) Its inputs
- d) Borrow bits

Answer: c

Explanation: Minuend and subtrahend are the two bits of input of a subtractor. If A and B are the two inputs of a subtractor then A is called minuend and B as subtrahend.

8. Full subtractor is used to perform subtraction of _____

- a) 2 bits
- b) 3 bits
- c) 4 bits
- d) 8 bits

Answer: b

Explanation: Full subtractor is used to perform subtraction of 3 bits, namely minuend bit, subtrahend bit and borrow from the previous stage. However, it also produces 2 outputs **BORROW** and **DIFFERENCE**.

9. The full subtractor can be implemented using _____

- a) Two XOR and an OR gates
- b) Two half subtractors and an OR gate
- c) Two multiplexers and an AND gate
- d) Two comparators and an AND gate

Answer: b

Explanation: A full subtractor has 3 input bits and two outputs bits **BORROW** and **DIFFERENCE**. The full subtractor can be implemented using two half subtractors and an OR gate.

10. The output of a subtractor is given by (if A, B and X are the inputs).

- a) $A \text{ AND } B \text{ XOR } X$
- b) $A \text{ XOR } B \text{ XOR } X$
- c) $A \text{ OR } B \text{ NOR } X$
- d) $A \text{ NOR } B \text{ XOR } X$

Answer: b

Explanation: The difference output of a subtractor is given by (if A, B and X are the inputs) $A \text{ XOR } B \text{ XOR } X$.

11. The output of a full subtractor is same as _____

- a) Half adder
- b) Full adder

- c) Half subtractor
- d) Decoder

Answer: b

Explanation: The sum and difference output of a full adder and a full subtractor are same. If A, B and C are the input of a full adder and a full subtractor then the output will be given by $(A \oplus B \oplus C)$, respectively.

4.) Procedure for the Design of Combinational Circuits

1. The basic building blocks of the arithmetic unit in a digital computers are _____

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- b) Adders
- c) Multiplexer
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- E. Write down the switching expression(s) for the output(s).

- a) B, C, D, E, A
- b) A, D, E, B, C
- c) A, B, E, C, D
- d) B, A, E, C, D

Answer: c

Explanation: Combinational circuits are the ones which do not depend on previous inputs and depends only on the present values. The given arrangement in option c is the right sequence for the designing of the combinational circuits.

6. All logic operations can be obtained by means of _____

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- b) OR and NOR operations

- c) OR and NOT operations
- d) NAND and NOR operations

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Explanation: Since, the logic gates NOR and NAND are known as universal logic gates, therefore it can be used to design all the three basic gates AND, OR and NOT. Thus, it means that any operations can be obtained by implementation of these gates.

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- c) Input, one
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Explanation: If the two numbers are unsigned, the bit conditions of interest are the output carry and a possible zero result.

9. If the two numbers include a sign bit in the highest order position, the bit conditions of interest are the sign of the result, a zero indication and _____

- a) An underflow condition
- b) A neutral condition
- c) An overflow condition
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Explanation: If the two numbers include a sign bit in the highest order position, the bit conditions of interest are the sign of the result, a zero indication and an overflow condition.

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- c) The total number of control lines
- d) All of the Mentioned

Answer: b

Explanation: In an ALU, status bit conditions are sometimes called condition code bits or flag bits. It is so called because they tend to represent the status of the respect flags after any operation.

5.) K-Map Simplification

1. Which statement below best describes a Karnaugh map?

- a) It is simply a rearranged truth table
- b) The Karnaugh map eliminates the need for using NAND and NOR gates
- c) Variable complements can be eliminated by using Karnaugh maps
- d) A Karnaugh map can be used to replace Boolean rules

Answer: a

Explanation: K-map is simply a rearranged truth table. It is a pictorial representation of truth table having a specific number of cells or squares, where each cell represents a Maxterm or a Minterm.

2. Which of the examples below expresses the commutative law of multiplication?

- a) $A + B = B + A$
- b) $A \cdot B = B + A$
- c) $A \cdot (B \cdot C) = (A \cdot B) \cdot C$
- d) $A \cdot B = B \cdot A$

Answer: d

Explanation: The commutative law of multiplication is $(A * B) = (B * A)$.

The commutative law of addition is $(A + B) = (B + A)$.

3. The Boolean expression $Y = (AB)'$ is logically equivalent to what single gate?

- a) NAND
- b) NOR
- c) AND
- d) OR

Answer: a

Explanation: If A and B are the input for AND gate the output is obtained as AB and after inversion we get $(AB)'$, which is the expression of NAND gate. NAND gate produces high output when any of the input is 0 and produces low output when all inputs are 1.

4. The observation that a bubbled input OR gate is interchangeable with a bubbled output AND gate is referred to as _____

- a) A Karnaugh map
- b) DeMorgan's second theorem
- c) The commutative law of addition
- d) The associative law of multiplication

Answer: b

Explanation: DeMorgan's Law: $\sim(P+Q) \Leftrightarrow (\sim P).(\sim Q)$ Also, $\sim(P.Q) \Leftrightarrow (\sim P)+(\sim Q)$.

5. The systematic reduction of logic circuits is accomplished by _____

- a) Symbolic reduction
- b) TTL logic
- c) Using Boolean algebra
- d) Using a truth table

Answer: c

Explanation: The systematic reduction of logic circuits is accomplished by using boolean algebra.

6. Each "1" entry in a K-map square represents _____

- a) A HIGH for each input truth table condition that produces a HIGH output
- b) A HIGH output on the truth table for all LOW input combinations
- c) A LOW output for all possible HIGH input conditions
- d) A DON'T CARE condition for all possible input truth table combinations

Answer: a

Explanation: Each "1" entry in a K-map square represents a HIGH for each input truth table condition that produces a HIGH output. Thus, it represents a minterm.

7. Each "0" entry in a K-map square represents _____

- a) A HIGH for each input truth table condition that produces a HIGH output
- b) A HIGH output on the truth table for all LOW input combinations
- c) A LOW output for all possible HIGH input conditions
- d) A DON'T CARE condition for all possible input truth table combinations

Answer: c

Explanation: Each "0" entry in a K-map square represents a LOW output for all possible HIGH input conditions. Thus, it represents Maxterm.

8. Which of the following statements accurately represents the two BEST methods of logic circuit simplification?

- a) Actual circuit trial and error evaluation and waveform analysis

- b) Karnaugh mapping and circuit waveform analysis
- c) Boolean algebra and Karnaugh mapping
- d) Boolean algebra and actual circuit trial and error evaluation

Answer: c

Explanation: The two BEST methods of logic circuit simplification are Boolean algebra and Karnaugh mapping. Boolean Algebra uses the Laws of Boolean Algebra for minimization of Boolean expressions while Karnaugh Map is a pictorial representation and reduction of the Boolean expression.

9. Looping on a K-map always results in the elimination of _____
- a) Variables within the loop that appear only in their complemented form
 - b) Variables that remain unchanged within the loop
 - c) Variables within the loop that appear in both complemented and uncomplemented form
 - d) Variables within the loop that appear only in their uncomplemented form

Answer: c

Explanation: Looping on a K-map always results in the elimination of variables within the loop that appear in both complemented and uncomplemented form.

10. Which of the following expressions is in the sum-of-products form?

- a) $(A + B)(C + D)$
- b) $(A * B)(C * D)$
- c) $A * B * (CD)$
- d) $A * B + C * D$

Answer: d

Explanation: Sum of product means that it is the sum of all product terms. Thus, the number is multiplied first and then it is added: $A * B + C * D$.

11. Which of the following is an important feature of the sum-of-products form of expressions?

- a) All logic circuits are reduced to nothing more than simple AND and OR operations
- b) The delay times are greatly reduced over other forms
- c) No signal must pass through more than two gates, not including inverters
- d) The maximum number of gates that any signal must pass through is reduced by a factor of two

Answer: a

Explanation: An important feature of the sum-of-products form of expressions in the given option is that all logic circuits are reduced to nothing more than simple AND and OR operations. Sum Of Product means it is the sum of product terms containing variables in complemented as well as uncomplemented forms.

12. Which of the following expressions is in the product-of-sums form?

- a) $(A + B)(C + D)$
- b) $(AB)(CD)$
- c) $AB(CD)$
- d) $AB + CD$

Answer: a

Explanation: $(A + B)(C + D)$ represents the product-of-sums form.

6.) Controlled Inverter

1. Controlled inverter is also known as _____

- a) Controlled buffer
- b) NOT gate
- c) Both controlled buffer and NOT gate
- d) Controlled gate

Answer: c

Explanation: Controlled inverter is also known as controlled buffer and NOT gate as well. It is used between output and a bus so that one can control whether the output is fed to the bus or not.

2. Why XOR gate is called an inverter?

- a) Because of the same input
- b) Because of the same output
- c) It behaves like a NOT gate
- d) It behaves like a AND gate

Answer: c

Explanation: The XOR (Exclusive Or) gate has a true output when the two inputs are different. When one input is true, the output is the inversion of the other. When one input is false, the output is the non-inversion of the other.

3. Controlled buffers can be useful _____

- a) To control the circuit's output into the bus
- b) In comparison of component's output with its input
- c) In increasing the output from its low input
- d) All of the Mentioned

Answer: a

Explanation: Controlled buffers can be useful when you have a wire (often called a bus) whose value should match the output of one of several components. By placing a

controlled buffer between each component output and the bus, you can control whether that component's output is fed onto the bus or not.

4. A logic circuit that provides a HIGH output for both inputs HIGH or both inputs LOW is

- a) Ex-NOR gate
- b) OR gate
- c) Ex-OR gate
- d) NAND gate

Answer: a

Explanation: EX-OR gate gives 1 if both inputs are different means 0 or 1 and gives 0 if both are same and EX-NOR is opposite of EX-OR gate, so it provides a HIGH output for both inputs HIGH or both inputs are LOW. Thus, EX-NOR produces output for even number of 1's or all 0s, while EXOR produces output for odd number of 1's.

5. What is the first thing you will need if you are going to use a macro-function?

- a) A complicated design project
- b) An experienced design engineer
- c) Good documentation
- d) Experience in HDL

Answer: d

Explanation: HDL stands for Hardware Description Language. In order to use a macro function, one needs to have experience in HDL for representing the structure and behaviour of digital circuits.

6. What is the major difference between half-adders and full-adders?

- a) Full-adders are made up of two half-adders
- b) Full adders can handle double-digit numbers
- c) Full adders have a carry input capability
- d) Half adders can handle only single-digit numbers

Answer: c

Explanation: Half adders have only two inputs A and B. When we add two 4 bit binary number like 0001 and 0011, then half adder can not be used because if the first bit of both the numbers is 1, then the sum would be 0 and carry would be 1. But this carry can not be added with the second bits addition of the number. So, half adders are useless. But in full adders, one more carry input is present, so that, if carry of one stage is present, it can be added with the next stage as it is done in normal addition. So, therefore, full adders have a carry input capability.

7. The binary subtraction of $0 - 0 = ?$

- a) Difference = 0, borrow = 0
- b) Difference = 1, borrow = 0
- c) Difference = 1, borrow = 1
- d) Difference = 0, borrow = 1

Answer: a

Explanation: The binary subtraction of $0 - 0 = 0$. Thus, it's difference is 0 as well as it's borrow.

8. How many basic binary subtraction operations are possible?

- a) 1
- b) 4
- c) 3
- d) 2

Answer: b

Explanation: 4 basic binary subtraction operations (0-0, 1-0, 0-1, 1-1) are possible.

$$0 - 0 = 0$$

$$0 - 1 = 1 \text{ (Borrow 1)}$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

9. When performing subtraction by addition in the 2's-complement system is?

- a) The minuend and the subtrahend are both changed to the 2's-complement
- b) The minuend is changed to 2's-complement and the subtrahend is left in its original form
- c) The minuend is left in its original form and the subtrahend is changed to its 2's-complement
- d) The minuend and subtrahend are both left in their original form

Answer: c

Explanation: When performing subtraction by addition in the 2's-complement system, the minuend is left in its original form and the subtrahend is changed to its 2's-complement. It is then added to the minuend. If the result has carry, then it's dropped and that's the final answer. Else, if the result has no carry, then the result is again converted to its 2's complement form and that's the final answer with a 'negative' sign.

10. What are the two types of basic adder circuits?

- a) Sum and carry
- b) Half-adder and full-adder
- c) Asynchronous and synchronous
- d) One and two's-complement

Answer: b

Explanation: There are two types of adder circuits: half-adder and full-adder. Half-Adder has 2 inputs while Full-Adder has 3 inputs. Whereas, both have two outputs SUM and CARRY.

11. Which of the following is correct for full adders?

- a) Full adders have the capability of directly adding decimal numbers
- b) Full adders are used to make half adders
- c) Full adders are limited to two inputs since there are only two binary digits
- d) In a parallel full adder, the first stage may be a half adder

Answer: d

Explanation: By using maximum of two half adders we can make a full adder for the first stage of a Parallel Full adder.

12. The selector inputs to an arithmetic/logic unit (ALU) determine the _____

- a) Selection of the IC
- b) Arithmetic or logic function
- c) Data word selection
- d) Clock frequency to be used

Answer: b

Explanation: An ALU performs basic arithmetic and logic operations and stores it in the accumulator. Examples of arithmetic operations are addition, subtraction, multiplication, and division. Examples of logic operations are comparisons of values such as NOT, AND and OR and any logical operations.

7.) 4-Bit Parallel Adder/Subtractor

1. For a 4-bit parallel adder, if the carry-in is connected to a logical HIGH, the result is _____

- a) The same as if the carry-in is tied LOW since the least significant carry-in is ignored
- b) That carry-out will always be HIGH
- c) A one will be added to the final result
- d) The carry-out is ignored

Answer: c

Explanation: For a 4-bit parallel adder, if the carry-in is connected to a logical HIGH, one will be added to the final result as a result because carry-in gives output as 1.

2. Fast-look-ahead carry circuits found in most 4-bit full-adder circuits which _____

- a) Determine sign and magnitude
- b) Reduce propagation delay

- c) Add a 1 to complemented inputs
- d) Increase ripple delay

Answer: b

Explanation: A carry-lookahead adder (CLA) is a type of adder used in digital logic. A carry-lookahead adder improves speed by reducing the amount of time required to determine carry bits. It reduces the propagation delay by making the hardware more complex. The ripple carry design is converted in such a way that carry over a group of bits of the adder becomes 2-level logic.

3. One way to make a four-bit adder to perform subtraction is by _____

- a) Inverting the output
- b) Inverting the carry-in
- c) Inverting the B inputs
- d) Grounding the B inputs

Answer: c

Explanation: A adder is a digital circuit which adds bits along with a carry bit from a previous stage, thus producing 2 outputs SUM and CARRY. Since, a four bit adder has four A, four B and a carry at the input end. So, for subtraction to be performed, all the Bs terminal should be inverted.

4. What distinguishes the look-ahead-carry adder?

- a) It is slower than the ripple-carry adder
- b) It is easier to implement logically than a full adder
- c) It is faster than a ripple-carry adder
- d) It requires advance knowledge of the final answer

Answer: c

Explanation: It is faster than ripple carry adder as it reduces the propagation delay by converting the ripple carry in such a way that the carry over a group of bits of the adder becomes 2-level logic.

5. Carry lookahead logic uses the concepts of _____

- a) Inverting the inputs
- b) Complementing the outputs
- c) Generating and propagating carries
- d) Ripple factor

Answer: c

Explanation: Look Ahead Carry Adder is a type of digital circuit which reduces the propagation delay. Carry lookahead logic uses the concepts of generating and propagating carries. Although in the context of a carry lookahead adder, it is most natural to think of

generating and propagating in the context of binary addition, the concepts can be used more generally than this.

6. What is one disadvantage of the ripple-carry adder?

- a) The interconnections are more complex
- b) More stages are required to a full adder
- c) It is slow due to propagation time
- d) All of the Mentioned

Answer: c

Explanation: The main disadvantage in using this type of adders is that the time delay increases as for each adder to add the carry should be generated in the previous adder, and for that to add the carry from the one before is required. However, this disadvantage is taken care of in Carry Look Ahead adder in which the ripple carry is converted in such a way that the carry over a group of bits of the adder becomes 2-level logic.

7. The carry propagation delay in 4-bit full-adder circuits _____

- a) Is cumulative for each stage and limits the speed at which arithmetic operations are performed
- b) Is normally not a consideration because the delays are usually in the nanosecond range
- c) Decreases in direct ratio to the total number of full-adder stages
- d) Increases in direct ratio to the total number of full-adder stages but is not a factor in limiting the speed of arithmetic operations

Answer: a

Explanation: A full adder is a digital circuit with 3 inputs and two outputs SUM and CARRY. The carry propagation delay in 4-bit full-adder circuits is cumulative for each stage and limits the speed at which arithmetic operations are performed.

8. What is Manchester carry chain?

- a) Is a chain of controlled inverter
- b) Variation of a carry-lookahead adder
- c) Variation of a full-adder
- d) Variation of a ripple carry adder

Answer: b

Explanation: The Manchester carry chain is a variation of the carry-lookahead adder that uses shared logic to lower the transistor count. However, the carry generating logic depends on the logic to generate the carries in the past.

9. The main disadvantage of Manchester carry chain is _____

- a) Ripple factor

- b) Propagation delay
- c) Capacitive load
- d) Both propagation delay and capacitive load

Answer: d

Explanation: Propagation delay is the measure of time taken by the output to go to the next state when the input is altered. One of the major downsides of the Manchester carry chain is that the capacitive load of all of these outputs, together with the resistance of the transistors causes the propagation delay to increase much more quickly than a regular carry lookahead.

10. The summing outputs of a half or full-adder are designated by which Greek symbol?

- a) Omega
- b) Theta
- c) Lambda
- d) Sigma

Answer: d

Explanation: The summing outputs of a half or full-adder are designated by “sigma” which is a Greek symbol. This same symbol is used to signify the Minterms in case of an SOP expression.

11. Why is a fast-look-ahead carry circuit used in the 7483 4-bit full-adder?

- a) To decrease the cost
- b) To make it smaller
- c) To slow down the circuit
- d) To speed up the circuit

Answer: d

Explanation: A Carry Look Ahead (CLA) Adder is a type of adder that reduce the propagation delay. A fast Carry Look Ahead Adder is more fast than a normal CLA. Since, it is easy to implement and can be implemented on any types of chip and have the capability to reduce propagation delay, which helps in increasing the speed of 7483 4-bit full-adder.

8.) Fast Adder & Serial Adder – 1

1. The inverter can be produced with how many NAND gates?

- a) 2
- b) 1
- c) 3
- d) 4

Answer: b

Explanation: The inverter can be produced with the help of single NAND gate, because we can send a single input twice through the same NAND gate together, thus producing the inverted version of the input as output. It works as an inverter.

2. One positive pulse with $t_w = 75 \mu s$ is applied to one of the inputs of an exclusive-OR circuit. A second positive pulse with $t_w = 15 \mu s$ is applied to the other input beginning $20 \mu s$ after the leading edge of the first pulse. Which statement describes the output's relation with the inputs?

- a) The exclusive-OR output is a 20 s pulse followed by a 40 s pulse, with a separation of 15 s between the pulses
- b) The exclusive-OR output is a 20 s pulse followed by a 15 s pulse, with a separation of 40 s between the pulses
- c) The exclusive-OR output is a 15 s pulse followed by a 40 s pulse
- d) The exclusive-OR output is a 20 s pulse followed by a 15 s pulse, followed by a 40 s pulse

Answer: d

Explanation: When both the input pulses are high or low X-OR output is low. But when one of the input is high and another is low or vice-versa, output is high. In this problem for the first 20uS one input is high and another is low. So, obviously output is a high. for next 15uS both the input is high so output is low and for remaining 40uS(75-20-15) first input is still high and second one is low so output is high.

3. How many NOT gates are required to implement the Boolean expression: $X = AB'C + A'BC$?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: a

Explanation: Since in the given expression two inputs are complemented. So, we require two NOT gate at the input. A NOT gate is a basic gate which accepts a single input and produces a single output, which is the inverted version of the input.

4. The carry look ahead adder is based on the principle of looking at the lower order bits of _____ and _____ if a high order carry is generated.

- a) Addend, minuend
- b) Minuend, subtrahend
- c) Addend, minuend
- d) Augend, addend

Answer: d

Explanation: The carry look ahead adder is based on the principle of looking at the lower order bits of the augend and addend if a high order carry is generated. A carry look ahead adder is a type of adder which reduces the propagation delay.

5. What are carry generate combinations?

- a) If all the input are same then a carry is generated
- b) If all of the output are independent of the inputs
- c) If all of the input are dependent on the output
- d) If all of the output are dependent on the input

Answer: b

Explanation: If the input is either 0, 0, 0 or 0, 0, 1 then the output will be 0 (i.e. independent of input) and if the input is either 1, 1, 0 or 1, 1, 1 then the output is 1 (i.e. independent of input). Such situation is known as carry generate combinations.

6. In serial addition, the addition is carried out _____

- a) 3 bit per second
- b) Byte by byte
- c) Bit by bit
- d) All bits at the same time

Answer: c

Explanation: In serial addition, the addition is carried out bit by bit.

7. How many shift registers are used in a 4 bit serial adder?

- a) 4
- b) 3
- c) 2
- d) 5

Answer: c

Explanation: There are two shift registers are used in a 4-bit serial adder, which is used to store the numbers to be added serially. Serial addition takes place bit by bit.

8. A D flip-flop is used in a 4-bit serial adder, why?

- a) It is used to invert the input of the full adder
- b) It is used to store the output of the full adder
- c) It is used to store the carry output of the full adder
- d) It is used to store the sum output of the full adder

Answer: c

Explanation: The D flip-flop, i.e. carry flip-flop, is used to store the carry output of the full

adder so that it can be added to the next significant position of the numbers in the registers.

9. What is ripple carry adder?

- a) The carry output of the lower order stage is connected to the carry input of the next higher order stage
- b) The carry input of the lower order stage is connected to the carry output of the next higher order stage
- c) The carry output of the higher order stage is connected to the carry input of the next lower order stage
- d) The carry input of the higher order stage is connected to the carry output of the lower order stage

Answer: a

Explanation: When the carry output of the lower order stage is connected to the carry input of the next higher order stage, such types of connection is called ripple carry adder in a 4-bit binary parallel adder.

10. If minuend = 0, subtrahend = 1 and borrow input = 1 in a full subtractor then the borrow output will be _____

- a) 0
- b) 1
- c) Floating
- d) High Impedance

Answer: b

Explanation: If minuend = 0, subtrahend = 1 and borrow input = 1 in a full subtractor then the borrow output will be 1. Because on subtracting 0 and 1, one borrow is taken and it proceeds till the next step (i.e $0 - 1 - 1 = 0$, borrow = 1).

11. A serial subtractor can be obtained by converting the serial adder by using the _____

- a) 1's complement system
- b) 2's complement system
- c) 10's complement
- d) 9's complement

Answer: b

Explanation: A serial subtractor can be obtained by converting the serial adder by using the 2's complement system. 9's complement and 10's complement are used for decimal numbers while adders deal with binary numbers.

12. The hexadecimal number (4B)16 is transmitted as an 8-bit word in parallel. What is the time required for this transmission if the clock frequency is 2.25 MHz?

- a) 444 ns
- b) 444 s
- c) 3.55 s
- d) 3.55 ms

Answer: a

Explanation: Because the clock pulse of 4-bit transmits the data of 8-bit word in parallel mode and this transmission is done at 2.25 MHz frequency. We know that: $f=1/t$ and we can find the time required for this transmission by the clock pulse.

Therefore, time = $(1/2.25) = 0.4444 \text{ us} = 444.44 \text{ ns} \sim 444\text{ns}$.

13. Internally, a computer's binary data are always transmitted on parallel channels which is commonly referred to as the _____

- a) Parallel bus
- b) Serial bus
- c) Data bus
- d) Memory bus

Answer: c

Explanation: A process consists of 3 types of buses: Control Bus, Data Bus and Address Bus. A computer's data is always in the binary form which is stored in the bus that transmits the data on any channels. It doesn't matter that it's in parallel or serial.

14. What is the frequency of a clock waveform if the period of that waveform is 1.25sec?

- a) 8 kHz
- b) 0.8 kHz
- c) 0.8 MHz
- d) 8 MHz

Answer: c

Explanation: By using the formula of frequency, we can find the frequency of clock waveform. Time period(t) of the waveform is = 1.25microseconds

$$f=1/t$$

Where 't' is the time taken by the clock waveform;

$$f=(1/1.25)$$

so, $f=0.8 \text{ MHz}$.

15. Why is parallel data transmission preferred over serial data transmission for most applications?

- a) It is much slower
- b) It is cheaper
- c) More people use it
- d) It is much faster

Answer: d

Explanation: Parallel data transmission preferred over serial data transmission for most applications because it is much faster as bits are transmitted simultaneously, whereas in serial data transmission, bits are transmitted one by one.

16. With surface-mount technology (SMT), the devices should _____

- a) Utilize transistor outline connections
- b) Mount directly
- c) Have parallel connecting pins
- d) Require holes and pads

Answer: b

Explanation: Surface-mount technology (SMT) is a method for producing electronic circuits in which the components are mounted or placed directly onto the surface of printed circuit boards (PCBs). An electronic device so made is called a surface-mount device (SMD). In the industry, it has largely replaced the through-hole technology construction method of fitting components with wire leads into holes in the circuit board. Both technologies can be used on the same board for components not suited to surface mounting such as large transformers and heat-sinked power semiconductors.

17. In most applications, transistor switches used in place of relays?

- a) They consume less power
- b) They are faster
- c) They are quieter and smaller
- d) All of the Mentioned

Answer: d

Explanation: Transistors are of less consuming power, faster, quieter, smaller and its implementation is too easy. So, in most applications transistor switches are more preferred. And also, transistors can be current-controlled or voltage-controlled depending on our choice.

18. What can a relay provide between the triggering source and the output that semiconductor switching devices cannot?

- a) Total isolation
- b) Faster
- c) Higher current rating

d) Total isolation and higher current rating

Answer: d

Explanation: A relay provides total isolation and higher current rating between the triggering source and the output that semiconductor switching devices cannot provide. This is why relays are used to drive high watt appliances at offices or other buildings.

19. The serial format for transmitting binary information uses _____

- a) A single conductor
- b) Multiple conductors
- c) Infrared technology
- d) Fiber-optic

Answer: a

Explanation: A conductor accepts the whole data and arranges it in a serial manner, which is transmitted as binary information. In serial transmission, data is transmitted bit by bit while in parallel transmission data is transmitted simultaneously.

20. Serial communication can be sped up by _____

- a) Using silver or gold conductors instead of copper
- b) Using high-speed clock signals
- c) Adjusting the duty cycle of the binary information
- d) Using silver or gold conductors instead of copper and high-speed clock signals

Answer: b

Explanation: For any serial data transmission there is required of continuously data supply and if the input supply (i.e. high speed clock signals) in a high amount the speed of serial communication can be increased. In serial communication, data is transmitted bit by bit. So the use of high speed clock pulses would make the process faster.

9.) BCD Adder

1. The decimal number system represents the decimal number in the form of _____

- a) Hexadecimal
- b) Binary coded
- c) Octal
- d) Decimal

Answer: b

Explanation: Binary-coded decimal (BCD) is a class of binary encodings of decimal numbers where each decimal digit is represented by a fixed number of bits, usually four or eight. Hexadecimal and Octal are number systems having base 16 and 8 respectively.

2. 2^9 input circuit will have total of _____

- a) 32 entries
- b) 128 entries
- c) 256 entries
- d) 512 entries

Answer: d

Explanation: 2^9 input circuit would have 512 ($2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 512$) entries.

3. BCD adder can be constructed with 3 IC packages each of _____

- a) 2 bits
- b) 3 bits
- c) 4 bits
- d) 5 bits

Answer: c

Explanation: Binary-coded decimal (BCD) is a class of binary encodings of decimal numbers where each decimal digit is represented by a fixed number of bits, usually four or eight. BCD adder can be constructed with 3 IC packages. Each of 4-bit adders is an MSI (Medium scale Integration) function and 3 gates for the correction logic need one SSI (Small Scale Integration) package.

4. The output sum of two decimal digits can be represented in _____

- a) Gray Code
- b) Excess-3
- c) BCD
- d) Hexadecimal

Answer: c

Explanation: The output sum of two decimal digits can be represented in BCD (Binary-coded decimal). Binary-coded decimal (BCD) is a class of binary encodings of decimal numbers where each decimal digit is represented by a fixed number of bits, usually four or eight.

5. The addition of two decimal digits in BCD can be done through _____

- a) BCD adder
- b) Full adder
- c) Ripple carry adder
- d) Carry look ahead

Answer: a

Explanation: The addition of two decimal digits in BCD can be done through BCD adder. Every input inserted, in addition by the user converted into binary and then proceed for

the addition. Whereas, Full Adder, Ripple Carry Adder and Carry Look Adder are for the addition of binary bits.

6. 3 bits full adder contains _____

- a) 3 combinational inputs
- b) 4 combinational inputs
- c) 6 combinational inputs
- d) 8 combinational inputs

Answer: d

Explanation: 3 bits full adder contains $2^3 = 8$ combinational inputs.

7. The simplified expression of full adder carry is _____

- a) $c = xy+xz+yz$
- b) $c = xy+xz$
- c) $c = xy+yz$
- d) $c = x+y+z$

Answer: a

Explanation: A full adder is a combinational circuit having 3 inputs and 2 outputs, namely SUM and CARRY. The simplified expression of full adder carry is $c = xy+xz+yz$.

8. Complement of F' gives back _____

- a) F'
- b) F
- c) FF
- d) FF'

Answer: b

Explanation: Complement means inversion. So, complement of F' gives back F , as per the Law of Involution.

9. Decimal digit in BCD can be represented by _____

- a) 1 input line
- b) 2 input lines
- c) 3 input lines
- d) 4 input lines

Answer: d

Explanation: Binary-coded decimal (BCD) is a class of binary encodings of decimal numbers where each decimal digit is represented by a fixed number of bits, usually four or eight. Decimal digit in BCD can be represented by 4 input lines. Since it is constructed with 4-bits.

10. The number of logic gates and the way of their interconnections can be classified as

-
- a) Logical network
 - b) System network
 - c) Circuit network
 - d) Gate network

Answer: a

Explanation: The number of different levels of logic gates is represented in a fashion which is known as a logical network.
